

Manan Pal

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Summary

Aspiring AI/ML Engineer with hands-on experience building and deploying end-to-end machine learning applications in NLP, computer vision, and graph neural networks. Skilled in Python, TensorFlow, PyTorch, Scikit-learn, and Streamlit with strong foundations in DSA, OOP, DBMS, and software development. Passionate about building intelligent systems focused on real-world usability, deployment, and interactive AI experiences.

Education

KIIT University — B.Tech in Computer Science and Engineering

May 2023 – May 2027

Technical Skills

Languages: Python, Java, C, SQL, JavaScript

AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision, Graph Neural Networks, Generative AI

Frameworks/Libraries: TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, PyTorch Geometric, NetworkX

Tools & Platforms: Git, GitHub, Streamlit, Jupyter Notebook, Google Colab, VS Code, Vercel, REST APIs

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Computer Networks, Graph Theory

Projects

Cybersecurity Attack Path Prediction using GNN | *Python · PyTorch · GraphSAGE · Streamlit* [GitHub](#)

- Built a synthetic graph-based cybersecurity dataset modeling network hosts and traffic behavior with node-level features.
- Trained GCN and GraphSAGE architectures for attack path prediction and node-level threat classification.
- Developed a complete pipeline with graph visualization and an interactive Streamlit dashboard for prediction analysis.

Brain Tumor Detection: CNN vs ViT with GAN Synthesis | *TensorFlow · MobileNetV2 · Vision Transformer · Streamlit* [GitHub](#)

- Developed MobileNetV2 CNN and Vision Transformer models for multi-class brain tumor classification on MRI scans.
- Integrated GAN-based synthetic MRI generation, Grad-CAM explainability, and NLP-based symptom analysis modules.
- Achieved 99.8% accuracy using Vision Transformer and deployed the solution with an interactive Streamlit interface.

NLP-Based Resume Ranking System | *Python · TF-IDF · Scikit-learn · Streamlit* [GitHub](#)

- Built an NLP-powered resume screening system that parses PDF resumes and ranks candidates using TF-IDF cosine similarity.
- Implemented skill-gap analysis highlighting matched and missing keywords with recruiter-style ranking outputs.
- Enhanced usability through automated PDF parsing, keyword extraction, and interactive Streamlit-based workflows.

Student Placement Readiness Prediction | *Scikit-learn · Random Forest · Tkinter* [GitHub](#)

- Trained and benchmarked multiple ML classifiers for placement readiness prediction using academic and extracurricular datasets.
- Achieved 89.5% accuracy using Random Forest and deployed the model with a Tkinter-based desktop interface.
- Provided probability-based outputs and feature importance visualization to improve interpretability and user interaction.

Developer Portfolio Website | *Next.js · TypeScript · Tailwind CSS · Vercel*

- Designed and deployed a modern portfolio website showcasing AI/ML projects, deployment workflows, and technical skills.
- Integrated GitHub, LinkedIn, and project deployment links to create a recruiter-focused professional presence.
- Implemented responsive UI components, smooth animations, and optimized layouts to improve recruiter engagement.

Virtual Internships

AICTE – EduSkills Virtual Internship Program

2025

AI/ML (AWS Academy) · Android Development (Google) · Generative AI Fundamentals

Certifications

AWS Cloud Foundations — AWS Academy

Machine Learning Foundations — AWS Academy

Ethical Decision Making in Tech — University of Colorado Boulder

Business for Good — London Business School